

SUPPLEMENTARY SEMESTER EXAMINATIONS – SEPTEMBER 2025

Program	: B.E. – Information Science and Engineering	Semester	: VI
Course Name	: Machine Learning	Max. Marks	: 100
Course Code	: 22IS62	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- What is Machine Learning? Discuss any two examples of machine learning applications. CO1 (06)
 - In a Medical Classification Model to diagnose if a patient is diabetic or not 50,000 patients were examined. The number of TP = 8000, FP = 5000, TN = 35,000, FN = 2000. Do the following for this Medical Classification Model:
 - Draw the Confusion Matrix
 - Calculate the Accuracy value
 - Assess the Sensitivity Value
 - Estimate the Specificity Value
 - Measure the Error Rate
 - Determine the Precision Value.
 - Explain the importance of the Gaussian distribution in machine learning. CO1 (07)
- Compare and contrast the following model evaluation strategies: CO1 (07)
 - Holdout Method
 - Random Subsampling.
 - In a Covid test of 1000 patients, there were 45 positive tests, of which 30 patients had covid and 15 were falsely tested positive. Of the 955 negative tests there were 5 that were incorrect, these patients had covid but were tested negatively. Draw the confusion matrix and calculate the accuracy, precision, recall, sensitivity and F1 score from the matrix. CO1 (07)
 - What is model overfitting, and how can cross-validation help in its detection? CO1 (06)

UNIT - II

- Find the dissimilarity matrix for the given dataset consisting of mixed attributes. CO2 (12)

Patient ID	Test-1 Nominal	Test-2 Ordinal	Test-3 Numeric
1	A	Excellent	45
2	B	Fair	22
3	C	Good	64
4	A	Excellent	28

- Given the following training examples, use the Find-S algorithm to determine the final hypothesis: CO2 (08)

Ex	Eyes	Nose	Head	Fcolor	Hair	Smile
1	Round	Triangle	Round	Purple	Yes	Yes
2	Square	Square	Square	Green	Yes	No
3	Square	Triangle	Round	Yellow	Yes	Yes
4	Round	Triangle	Round	Green	No	No
5	Square	Square	Round	Yellow	Yes	Yes

4. a) Apply candidate elimination algorithm for the given training samples and find the Final hypothesis. CO2 (10)

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Sunny	Warm	High	Strong	Warm	Same	Yes
3	Rainy	Cold	High	Strong	Warm	Change	No
4	Sunny	Warm	High	Strong	Cool	Change	Yes

- b) Discuss the need of dimensionality reduction in machine learning models and Provide step-by-step procedure of LDA algorithm. CO2 (10)

UNIT - III

5. a) Define the concept of a maximum margin hyperplane in the context of linear SVMs. CO3 (06)
- b) Consider the data set given below where 'Y' is the response variable and 'X' is predictor variable. Obtain the simple linear regression equation. CO3 (06)

X	95	85	80	70	60
Y	85	95	70	65	70

- c) Apply Naïve Bayes Classifier for the given dataset and predict the outcome for the feature {Yellow, Sports, Imported} CO3 (08)

Color	Type	Origin	Stolen
Red	Sports	Domestic	Yes
Red	Sports	Domestic	No
Red	Sports	Domestic	Yes
Yellow	Sports	Domestic	No
Yellow	Sports	Imported	Yes
Yellow	SUV	Imported	No
Yellow	SUV	Imported	Yes
Yellow	SUV	Domestic	No
Red	SUV	Imported	No
Red	Sports	Imported	Yes

6. a) Write and explain the Classification Tree Algorithm. Describe the concept of 'Gini Index' as an attribute selection method. CO3 (06)
- b) For the given sample dataset of COVID-19 infection, apply ID3 algorithm to select the best feature to find whether a person is infected or not. CO3 (08)

ID	Fever	Cough	Breathing issues	Infected
1.	No	No	No	No
2.	Yes	Yes	Yes	Yes
3.	Yes	Yes	No	No
4.	Yes	No	Yes	Yes
5.	Yes	Yes	Yes	Yes
6.	No	Yes	No	No
7.	Yes	No	Yes	Yes
8.	Yes	No	Yes	Yes
9.	No	Yes	Yes	Yes
10.	Yes	Yes	No	Yes
11.	No	Yes	No	No
12.	No	Yes	Yes	Yes
13.	No	Yes	Yes	No
14.	Yes	Yes	No	No

- c) Illustrate Support Vector Machine for linearly separable and inseparable data. CO3 (06)

UNIT- IV

7. a) Apply **k-means** clustering algorithm with Euclidean distance measure to Cluster the following 8 data points A1(2,10), A2(2,5), A3(8,4), A4(5,8), A5(7,5), A6(6,4), A7(1,2), A8(4,9), Initial centroids are data points A1, A4 and A7. CO4 (10)
- b) Using **AGNES** cluster the data points A (1,1), B (1.5,1.5), C (5,5), D (3,4), E (4,4), F (3,3.5). Use Euclidean distance. Draw the Dendrogram **and** the Clustering Hierarchy in X-Y space. CO4 (10)
8. a) Apply **k-means** clustering algorithm with Euclidean distance measure to Cluster the following 8 data points A1(2,10), A2(2,5), A3(8,4), A4(5,8), A5(7,5), A6(6,4), A7(1,2), A8(4,9), Initial centroids are data points A1, A4 and A7. CO4 (10)
- b) Given the dataset and distance matrix in the below table, apply divisive analysis clustering(**DIANA**) algorithm to form clusters. CO4 (10)

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

UNIT - V

9. a) Consider the two dimensional patterns (2, 1),(3, 5),(4, 3),(5, 6), (6, 7),(7, 8). Compute the principal component using PCA Algorithm. CO5 (10)
- b) What do you understand by the term 'Linear Separability'? Illustrate graphs which depicts linear and Non-Linear Separability. CO5 (04)
- c) Explain how do AdaBoost and Random Forest handle weak learners. CO5 (06)
10. a) Explain the steps involved in the Principal Component Analysis (PCA) algorithm and how it is used for dimensionality reduction. CO5 (08)
- b) Differentiate the ensemble concept of Bagging and Boosting to make a collection of prediction. CO5 (06)
- c) What is a Perceptron? Illustrate a Perceptron for AND function with a diagram. CO5 (06)
